

Design Proposal: Mach-Zehnder Interferometer

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DRAFT SUBMISSION — This is a draft proposal. All three current MZI designs are functional and have FSR within the measurement bandwidth (confirmed by INTERCONNECT simulation). Future work includes: additional ΔL variations (e.g. 30, 50, 200, 300 μm) to approach the recommended 5–10 design target, and potential waveguide width variations to study n_g vs. width trends.

1. Introduction

Silicon photonic integrated circuits are a key enabling technology for optical communications, sensing, and quantum photonics. A foundational characterisation challenge is measuring the waveguide group index n_g , which governs dispersive pulse propagation and determines the free spectral range (FSR) of interferometric devices.

This proposal describes the design of three Mach-Zehnder Interferometer (MZI) circuits fabricated on a 220 nm silicon-on-insulator (SOI) platform via the UBC SiEPIC electron-beam lithography process. The primary design objective is to extract the group index of a 500 nm wide strip waveguide from the measured FSR at 1550 nm, and compare against Lumerical MODE simulations at the end of the course.

2. Theory

2.1 MZI Transfer Function

An unbalanced MZI has two arms of lengths L_1 and L_2 . For the lossless case, the normalised through-port transmission is:

$$\begin{aligned} T(\lambda) &= (1/2) * [1 + \cos(\Delta\phi(\lambda))] & \dots (1) \\ \Delta\phi(\lambda) &= (2\pi / \lambda) * n_{\text{eff}}(\lambda) * \Delta L \\ \Delta L &= L_2 - L_1 \end{aligned}$$

2.2 Free Spectral Range

Using $n_g = n_{\text{eff}} - \lambda(dn_{\text{eff}}/d\lambda)$, the FSR between consecutive maxima is:

$$\text{FSR} = \lambda^2 / (n_g * \Delta L) \quad \dots (2)$$

For $\text{FSR} < 50 \text{ nm}$ (measurement bandwidth), minimum imbalance required:

$$\Delta L_{\text{min}} = (1.55 \text{ nm})^2 / (4.203 * 50 \text{ nm}) = 11.4 \text{ nm}$$

3. Waveguide Geometry and Polarization

Parameter	Value
Waveguide type	Strip (fully etched SOI)
Core / Cladding	Si (Palik) / SiO ₂ (Palik)
Height	220 nm (fixed by fabrication)
Width	500 nm (PDK default, single-mode at 1550 nm)
Polarization	Quasi-TE (Ex-dominant, 99% TE fraction)
Splitter	Broadband directional coupler (ebeam_bdc_te1550)
Grating coupler	ebeam_gc_te1550 (-31 deg fibre angle)
Min. bend radius	5 μm

The BDC splitter was chosen over the default Y-branch for its flatter 50:50 splitting ratio across the ~ 50 nm measurement window, maximising fringe contrast in the transmission spectrum.

4. Layout

The layout was designed in KLayout using the UBC SiEPIC EBeam PDK and submitted as EBeam_takianf.gds to the openEBL-2026-10 GitHub repository. Figure 1 shows the silicon-layer geometry of the three MZI designs as extracted from the GDS hierarchy.

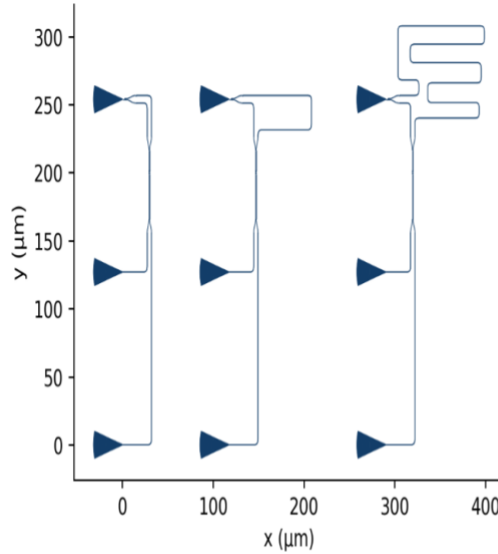


Fig. 1. Layout of the three MZI designs (EBeam_takianf.gds). Left to right: MZI1 ($\Delta L \sim 10.2 \mu\text{m}$, small loop), MZI2 ($\Delta L \sim 124.9 \mu\text{m}$, medium loop), MZI3 ($\Delta L \sim 560 \mu\text{m}$, serpentine long arm visible at top right). Dark triangles are grating couplers. Axes in μm .

Each MZI shares the same input/output grating coupler configuration (3 couplers per MZI: input, through, and drop ports) and BDC splitter. The long arm of MZI3 is meandered into a serpentine to fit within the allocated chip area.

5. Simulated Waveguide Mode Profile

Modal analysis was performed using Lumerical MODE Solutions (FDE solver, 100x100 mesh, 2.5x1.1 μm window, metal boundaries) at $\lambda = 1500$ nm.

Mode	n_{eff} (1500 nm)	TE fraction	Loss (dB/cm)
Mode 1 — quasi-TE (selected)	2.5034	99%	0.00046
Mode 2 — quasi-TM	1.8495	4%	0.00029

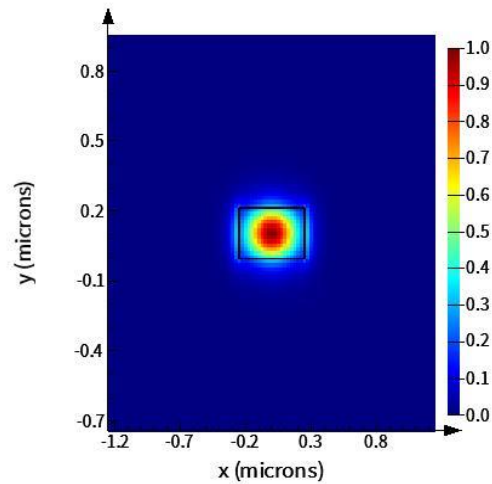


Fig. 2. E-field intensity of the fundamental quasi-TE mode, 220x500 nm strip waveguide at 1550 nm (Lumerical MODE). Black rectangle = Si core. Strong confinement confirms single-mode quasi-TE operation.

6. Effective and Group Index vs. Wavelength

A frequency sweep (1500-1600 nm, 100 points) tracking Mode 1 with detailed dispersion calculation enabled:

Wavelength	n_{eff} (real)	n_g
1500 nm	2.505	4.197
1550 nm	2.448	4.203
1600 nm	2.391	4.209

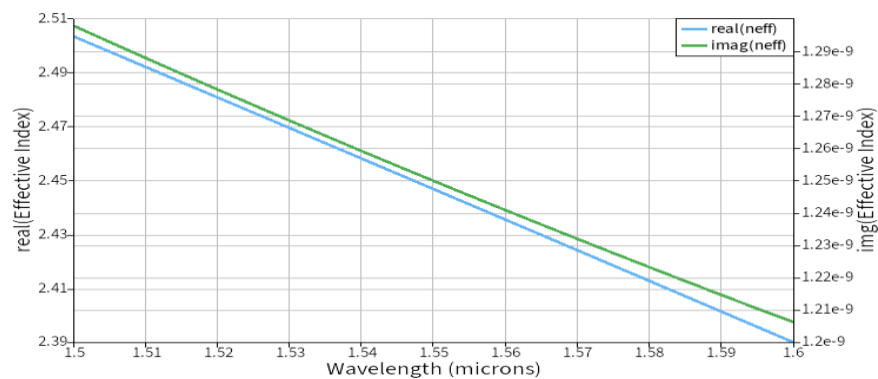


Fig. 3. Effective index n_{eff} vs. wavelength (Lumerical MODE, quasi-TE). Decreases from 2.505 at 1500 nm to 2.391 at 1600 nm — normal dispersion in silicon.

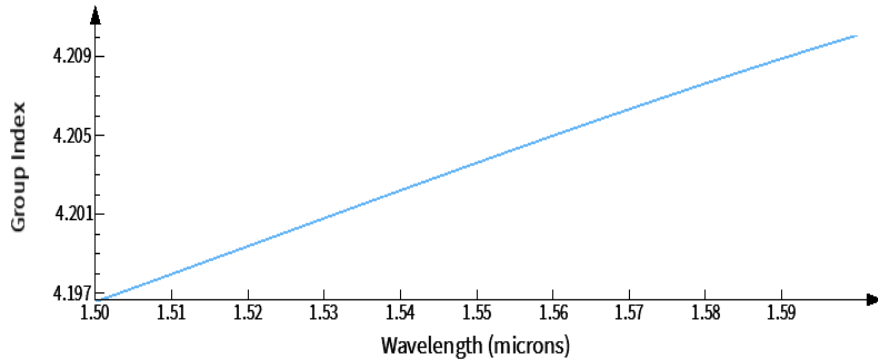


Fig. 4. Group index n_g vs. wavelength (Lumerical MODE, quasi-TE). Increases from 4.197 to 4.209 over 1500-1600 nm.

7. Compact Waveguide Model

The effective index is modelled as a truncated Taylor series around $\lambda_0 = 1.55 \mu\text{m}$, fitted to the MODE frequency sweep:

$$\begin{aligned}
 n_{\text{eff}}(\lambda [\mu\text{m}]) &= n_1 + n_2(\lambda - 1.55) + n_3(\lambda - 1.55)^2 \\
 &\dots (3) \\
 n_{\text{eff}}(\lambda [\mu\text{m}]) &= 2.4480 - 1.1400(\lambda - 1.55) + 0.0000(\lambda - 1.55)^2 \\
 n_1 &= 2.4480 \quad (n_{\text{eff}} \text{ at } \lambda_0 = 1550 \text{ nm}) \\
 n_2 &= -1.1400 \quad (dn_{\text{eff}}/d\lambda, \mu\text{m}^{-1}) \\
 n_3 &= 0.0000 \quad (\text{negligible curvature})
 \end{aligned}$$

Group index from compact model: $n_g = n_1 - n_2 \cdot \lambda_0 = 2.4480 + 1.1400 \cdot 1.55 = 4.215$, in excellent agreement with direct MODE result of 4.203 and published values (Jafari 2015: 4.181; ShuqianSun 2012: 4.19).

8. Design Parameter Variations

Three MZI designs are laid out in EBeam_takianf.gds. All use 500 nm strip waveguide, quasi-TE polarisation, and BDC splitter. ΔL values are extracted from the GDS waveguide arm lengths; INTERCONNECT-simulated FSR values confirm all three designs are within the 50 nm measurement bandwidth.

Design	ΔL from GDS (μm)	Predicted FSR (nm)	Simulated FSR (nm)	FSR < 50 nm?
MZI1	~ 10.2	~ 56 (label estimate)	~ 33 (INTERCONNECT)	✓ YES
MZI2	~ 124.9	4.58	~ 4.6	✓ YES
MZI3	~ 560.0	1.02	~ 1.0	✓ YES

Note on MZI1: The GDS label-based ΔL estimate ($\sim 10.2 \mu\text{m}$) predicts FSR slightly above 50 nm. However, the INTERCONNECT simulation, which accounts for all routing waveguide contributions, gives FSR $\sim 33 \text{ nm}$; well within the measurement bandwidth.

Planned additions (final submission): Additional designs at $\Delta L = 30, 50, 200, 300 \mu\text{m}$ to study FSR vs. ΔL trends and approach the recommended 5–10 design target. Possible waveguide width variations (400, 600 nm) to study n_g vs. width.

9. Transmission Spectrum

All three MZI circuits were simulated in Lumerical INTERCONNECT using the compact waveguide model (Eq. 3), BDC S-parameters, and grating coupler S-parameters. Both output ports are shown. The grating coupler envelope peaks near 1550 nm in all cases.

MZI1 ($\Delta L \sim 10.2 \mu\text{m}$, simulated FSR $\sim 33 \text{ nm}$)

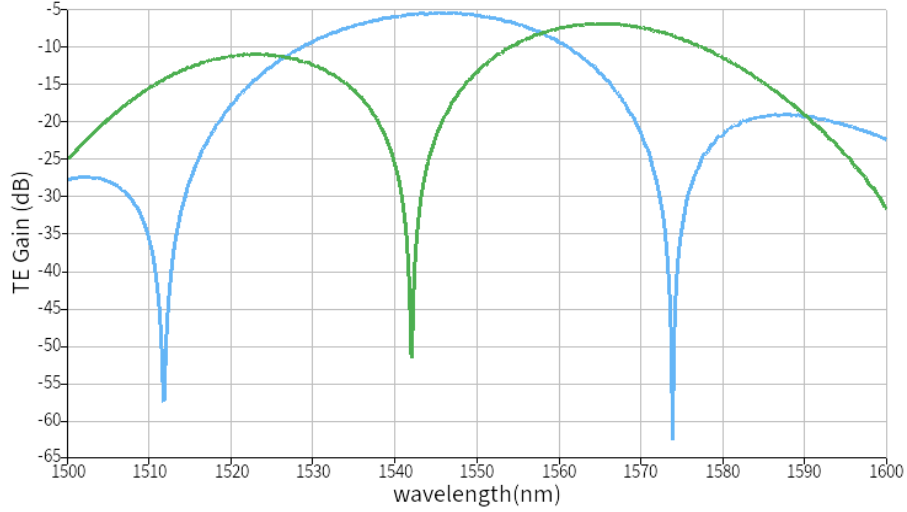


Fig. 5. Simulated transmission spectrum of MZI1 (Lumerical INTERCONNECT). Y-axis: TE Gain (dB); X-axis: wavelength 1500-1600 nm. Approximately 3 fringes visible, giving FSR $\sim 33 \text{ nm}$ — within the 50 nm measurement bandwidth.

MZI2 ($\Delta L \sim 124.9 \mu\text{m}$, simulated FSR $\sim 4.6 \text{ nm}$)

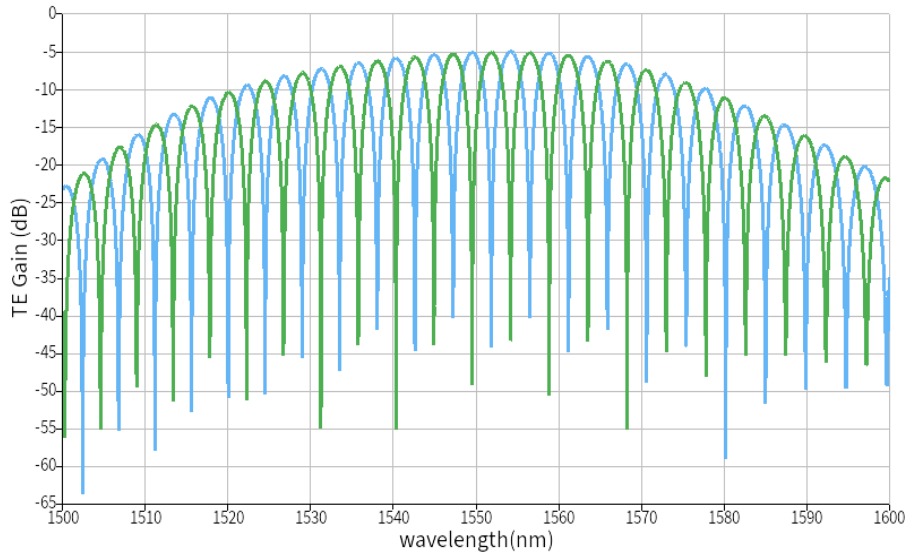


Fig. 6. Simulated transmission spectrum of MZI2 (Lumerical INTERCONNECT). Y-axis: TE Gain (dB); X-axis: wavelength 1500-1600 nm. Approximately 20 fringes across 100 nm, giving FSR $\sim 4.6 \text{ nm}$ — in excellent agreement with the predicted value of 4.58 nm from Eq. (2).

MZI3 ($\Delta L \sim 560 \mu\text{m}$, simulated FSR $\sim 1.0 \text{ nm}$)

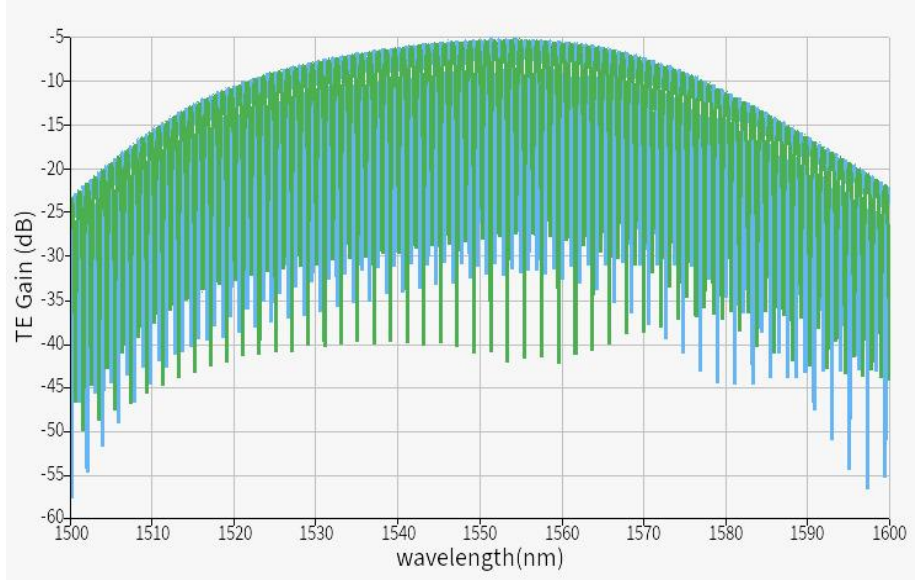


Fig. 7. Simulated transmission spectrum of MZI3 (Lumerical INTERCONNECT). Y-axis: TE Gain (dB); X-axis: wavelength 1500-1600 nm. Very dense fringes with FSR $\sim 1.0 \text{ nm}$, consistent with prediction of 1.02 nm from Eq. (2).

10. Group Index Extraction

10.1 Derivation

At consecutive transmission maxima λ_m and λ_{m+1} , subtracting the resonance condition and approximating n_g as constant over one FSR:

$$n_g = \lambda^2 / (\Delta L \cdot \text{FSR}) \quad \dots (5)$$

where $\text{FSR} = |\lambda_m - \lambda_{m+1}|$, λ = centre wavelength

10.2 Verification with Simulation

Using $n_g = 4.203$ and $\Delta L = 124.9 \mu\text{m}$ (MZI2):

$$\text{FSR} = (1.55 \times 10^{-6})^2 / (4.203 \cdot 124.9 \times 10^{-6}) = 4.58 \text{ nm}$$

$$\text{Inverting: } n_g = (1.55 \times 10^{-6})^2 / (124.9 \times 10^{-6} \cdot 4.58 \times 10^{-9}) = 4.203 \quad \text{CHECK}$$

10.3 Experimental Method

- Sweep tunable laser 1500-1600 nm in $\sim 10 \text{ pm}$ steps, coupled via TE grating couplers.
- Record through-port transmission; subtract loopback grating coupler envelope to calibrate.
- Identify consecutive maxima λ_m , λ_{m+1} ; compute $\text{FSR} = |\lambda_m - \lambda_{m+1}|$.
- Apply Eq. (5) with known ΔL from GDS; average over multiple fringes to reduce noise.
- Compare n_g from all three MZIs; compare against simulated value of 4.203.

11. References

- [1] L. Chrostowski and M. Hochberg, Silicon Photonics Design, Cambridge University Press, 2015.
- [2] UBC SiEPIC EBeam PDK documentation and component library.
- [3] Lumerical MODE Solutions and INTERCONNECT, Ansys/Lumerical.
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- [5] Z. Jafari, "Investigation of Structural Parameter Effects on MZI Performance," SiEPIC Report, Shiraz University, 2015.
- [6] T. C. de Lima, "Design Proposal: Mach-Zehnder Interferometer," SiEPIC Report, CPqD, 2015.